Fluorine

9 **F** The most reactive of all halogens, fluorine has a reputation as a dangerously unstable element. It reacts violently with metals, hydrogen gas, and water and will even react with some noble gases, which are normally inert.

Fluoride

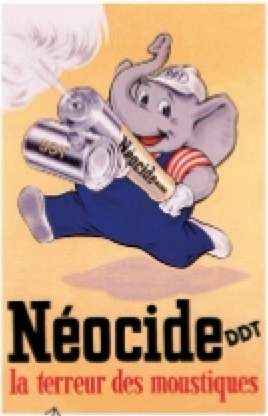
Some fluorine compounds play a vital role in oral health. When added to toothpaste and the water supply in the form of safe fluoride, these compounds help prevent tooth decay by strengthening tooth enamel.

Fluorite

Also known as fluorspar, fluorite is a naturally occurring mineral consisting mainly of the compound calcium fluoride. When fluorite reacts with sulfuric acid, it produces hydrogen fluoride, which is the main source of fluorine in industry.

Fluorite crystals

Green color is due to impurities in the crystal



This poster from the 1950s shows an elephant spraying a can of DDT to kill mosquitoes.

Deadly pesticide

In the 1940s and 1950s, the chlorine-based chemical DDT (dichloro-diphenyl-trichloroethane) was used widely as a pesticide to curb insect-borne diseases. However, concerns about its impact on wildlife and human health grew until it was banned in the US in 1972. Today, it is banned in most countries and severely restricted in others.

These strong pipes—for use in home construction—are composed of PVC.

Indispensable plastic

Arguably one of the most important compounds to contain chlorine is polyvinylchloride (PVC). This type of plastic has a huge number of uses in and around the home, including water pipes, lawn furniture, and waterproof clothing.

British World War II gas mask

Filter unit absorbs deadly chlorine gas.

Gas warfare

A pungent, yellow-green gas, chlorine affects the eyes, nose, throat, and lungs by causing a choking and irritating effect that can lead to suffocation. Chlorine was first used as a chemical weapon in World War I, which led to the development of gas masks for troops. By World War II, gas masks were provided for troops and civilians alike.

Chlorine-based bleach

Bleaching agent

Paper products such as napkins, tissues, and printer paper can be made bright white by using chlorine as a bleaching agent. Household bleach typically contains a chlorine compound called sodium hypochlorite. It destroys germs by breaking down their cell walls.